Temperature & Humidity Monitoring for Poultry Farms using IOT

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Abstract - Malaysian society consumes 49.30 kg of chicken and 370 eggs as a source of protein each year and in 2020. Malaysia has over 3000 poultry farms across the country. Nowadays, poultry farms have modernized tools using Internet-of-Things to ease accessibility. Internet-of-things supports poultry farmers to preserve the chicken cage environment. High environmental temperatures destructively influence the health of chickens; unstable humidity levels affect the chicken quality and weight in the cage. Improper collection and analysis of cage environmental data delays the appropriate action to be taken. Therefore, designing a web-based application is essential to assist farmers and engineers in managing the chicken poultry farms to collect and analyze the data necessary to monitor the chicken health and quality. In this paper, a mechanism has been developed to monitor the temperature and humidity in the poultry cage. DHT22 sensors were used to measure the temperature and humidity levels of the cage. The readings from the sensors were processed by the microcontroller and automatically and subsequently sent to a central database, where the user can monitor and update the farm's status. Based on the temperature and humidity values received from the sensors, appropriate action will be taken by the system to turn on the lamp and the fan to stabilize the chicken's cage. Once the temperature and humidity cross a threshold, a notification will be sent to alert the situation of the cage to the farmer.

Keywords- Poultry Farm, DHT22 Sensor, Temperature, Humidity, Monitor, Controlling, Raspberry Pi 4.

I. INTRODUCTION

The poultry industry in Malaysia is an essential agriculture sector since it supplies Malaysians with a low-cost source of protein. According to statistics by Hirschmann [1], Malaysians with an affordable source of protein consume about 49.30 kg of chicken meat and 370 eggs each year. Chicken is the most popular meat in Malaysia, and it is provided by over 3000 poultry farms across the country. Around 1,716 chickens are exported to Singapore for food needs, making Malaysia a large supplier of chicken exports to Singapore [2]. Poultry farmers in Malaysia have an important role in maintaining the health of poultry. In addition to monitoring the health of poultry, farmers also need to monitor the temperature of the poultry house. A good temperature for a cage does not exceed 37°C [3]. In the modern era, poultry farms have modernized tools using the Internet-of-things concept to make it easier for people to work. The Internet-of-things helps small poultry farmers ease work and maintain the chicken cage environment [4]. Many of the small poultry farmers in Malaysia still use old technology to manage the poultry farm, because of the high prices of advanced technology for poultry farms [5]. For this reason, the research involves monitoring the temperature of the chicken cages using the Internet-of-things (IoT) to help small farmers. The project developed a temperature sensor to monitor the temperature in the chicken cage and control the temperature of the chicken cage to avoid significant changes. Modernized chicken farmers use advanced technology in managing poultry farms and using closed housing systems. Also, modern technology helps farmers manage the chicken cage automatically, so they don't need to go to the chicken cage [6].

Ali and Rahman (2020) found that the health of chickens is very sensitive to high environmental temperatures. Chickens need a conducive environment and a suitable temperature to avoid being exposed to chicken diseases [7]. According to Choosumrong, the poultry environment using a traditional method can never be maintained accurately because of the lack of monitoring by the farmer effecting the humidity in the chicken quality and weight [8]. According to Zeyad et al., lack of control and monitoring of the chicken environment could restrict the outcome of the production [9]. Batuto et al. found that improper collection and analysis of cage environmental data delays the appropriate action from being taken. In this case, the temperature and humidity environment result will be critical in data processing to ensure stability [10].

The significance of the proposed research is that it monitors the temperature and humidity of the chicken cages in the poultry farm. Therefore, designing a web-based application is necessary to help farmers and engineers manage chicken poultry farms and collect and analyze the data to control chicken health and quality.

II. Literature Review

In traditional poultry farms, the farmer directly monitors the temperature in the cage and checks the condition of the environment cage. Traditional poultry farms will manually turn on the lights as a chicken heater to control the temperature and humidity in the chicken cage. In a traditional poultry farm, to determine the environmental condition of chickens, the chicken farmers can see directly by observing the environment of the chicken cage. Chicken farmers need to enter the cage to monitor the chickens' environment [10][11].

In the modern era, implementing IoT in poultry farms helps chicken farmers to monitor [13]. In system Goswami [14], monitoring system using sensor from DHT22 as the temperature and humidity sensor, monitor serial shows temperature and humidity values reported by the sensor.

In poultry farming, controlling livestock is necessary to develop the productivity and quality of chickens. Healthy and well-maintained chickens produce quality poultry meat. In the traditional poultry farming technique, the farmer needs to control the environment of chicken cages such as temperature & humidity of environment [15].

Chicken farmers are easier controlled by automation with smart poultry farm. A smart poultry system needed sensors to control the environment of the chicken cage, such as temperature and humidity [16]. Automated control by using a device can control the level of temperature and humidity in the chicken cage [17].

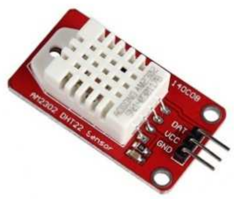

Fig. 1: Temperature and Humidity Sensor (DHT22)

Smart poultry farm using many sensors to develop function and feature, such as temperature and humidity sensors. DHT22 sensor depicts in fig. 1 measures temperature and humidity levels. Result measurement from the sensors will be processed by the microcontroller, the result automatically sent to a website to the user monitor and update the farm's status [18].

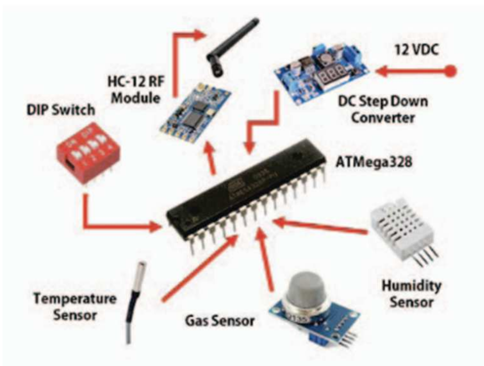

Fig. 2: Block Diagram for smart poultry farm [19].

The related products is block diagram from Ahmad Ammar et al. Ahmad Ammar et al. used 4 slave module and a master module separated between output system for activating sensor and fan, the picture on fig. 2 is slave module consists of ATMega 328 as micro controller, DIP switch, HC-1 RF, DC step down converter, Humidity Sensor, Gas sensor, Temperature Sensor [19].

A. *Preliminary study using DHT11 Sensor, Raspberry Pi 3 Model B and IOTA Blockchain in Poultry Farm*

System development by Elham et al. study are using DHT11 & Raspberry Pi 3 Model B depicts in fig. 4 as hardware requirement and integrated with IOTA Blockchain in Elham et al. system. The disadvantage of their system is using DHT 11 and the latest of module is DHT22 [11]. The fig. 3 depicts block diagram from Elham et al. study

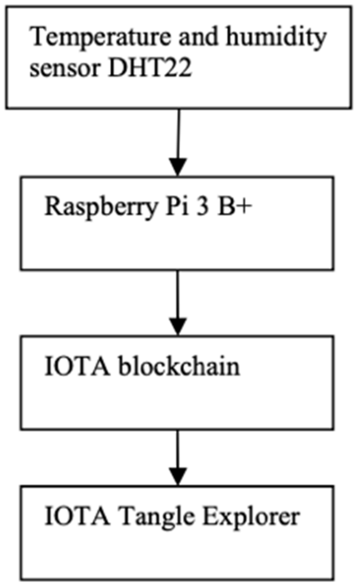

Fig. 3: Block Diagram from Elham et al study [11].

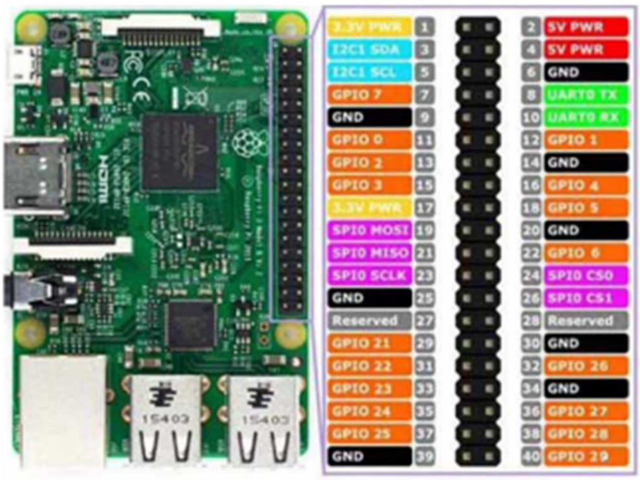

Fig. 4: Raspberry PI 3 model B [11].

B. *Temperature control using ZigBee modul as Wireless Sensor Network (WSN)*

In a study by Bilal Ghazal et al., Design and Analysis Automatic Temperature Control in The Broiler Poultry Farm Based on Wireless Sensor Network system, the experiment creating a temperature and humidity monitoring system with the ZigBee method as wireless connectivity modules. ZigBee module is an IEEE 802.15.4 based module allows devices to communicate wirelessly with another sensor. ZigBee serially connected to the server connected to PIC microcontroller as a module for measuring temperature and humidity in this study [16].

Based on Bilal Ghazal system, creating temperature and humidity monitoring system with the ZigBee method as wireless connectivity modules. ZigBee module allows devices to communicate wirelessly with another sensor and PIC Microcontroller was used as microcontroller and module for measure temperature and humidity. Disadvantages of their system are not provided application or software. Also, the system needs a wireless module to connect the microcontroller to server.

C. Smart Farm: Extending Automation To The Farm System

In Azlin et al study used a slave module and a master module using ATMega 328 as microcontroller, ESP8266 as Wi-Fi module to connect wireless microcontroller and server depicts in fig. 6. Flowchart of sensor test based Azlin et al study depicts in fig. 5. Disadvantages of the system is too many energies consumption due many sensors used. The system used 2 sensors for temperature and humidity since DHT22 can used as temperature and humidity sensor accurately [20] .

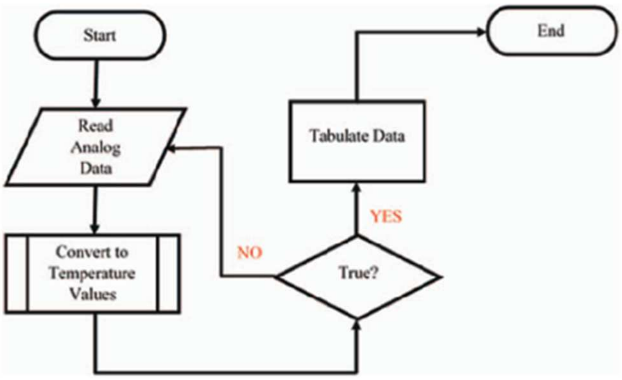

Fig. 5: Flowchart of sensor test based Azlin et al study [20].

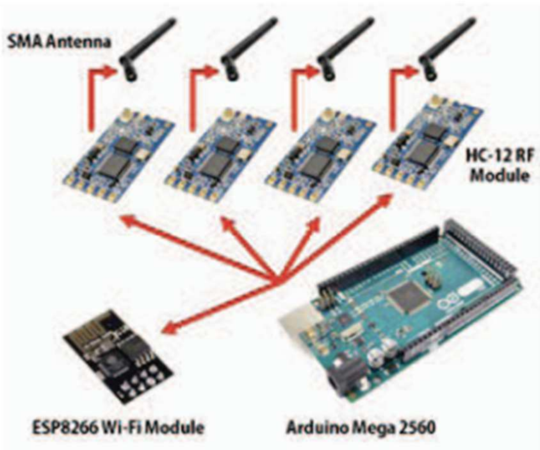

Fig. 6: Communication module to Master module by Azlin et al [20].

III. Methodology

This section describes the methods used, such as specific research procedures, materials, system design and modules used. This study applied agile methodology, which types Agile Unified Process or known as AUP. Agile Unified Process is used to develop and design the proposed system. Agile Unified Process implemented to find user needs and describe development techniques, show development techniques, determine how the system works through diagrams and tools used to develop the plan. There are 4 phases involved in Agile Methodology Process consists of inception, elaboration, construction, and transition. In fig. 7 depicts the class diagram for smart poultry farm system.

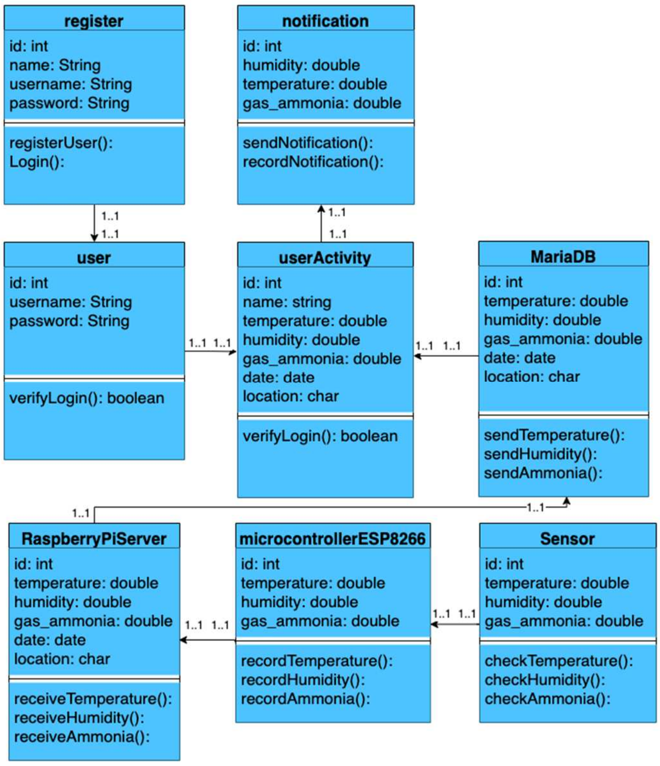

Fig. 7: Class diagram smart poultry farm system

The methodology used to develop this project is Agile Unified Process methodology. In AUP, there will be an interaction between the user and the engineer to know the software needed in every phase and the proximity between the customer (target user) and developer to build system. Advance in the development of the system because assisted by a tester (target user/chicken farmer) to check the system, the developer evaluated if there is an error.

Fig. 8: Raspberry Pi 4 Model B

Fig. 8 depicts the main purpose of making the Raspberry Pi 4 B Model is to encourage learning and experimentation. Raspberry PI 4 Model B has a RAM capacity of 512 MB to run the system. The Raspberry PI 4 Model B uses an ARM11 processor, the most potent processor currently used in microcomputers [21]. The function of Raspberry Pi as a server to store the data to provide Tier 3 of an IoT Architecture.

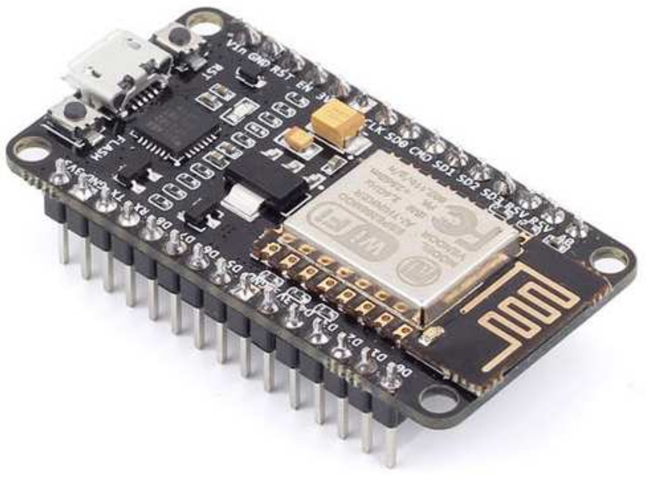

Fig. 9: ESP8266

Node MCU or ESP8266 shown in fig. 9 is a microcontroller function as a controller for the DHT22 sensor and sends the results of sensor readings through a router to be received by the Raspberry Pi. Python Language is needed to help communication between the ESP8266 and the Raspberry Pi 4 Model B [22].

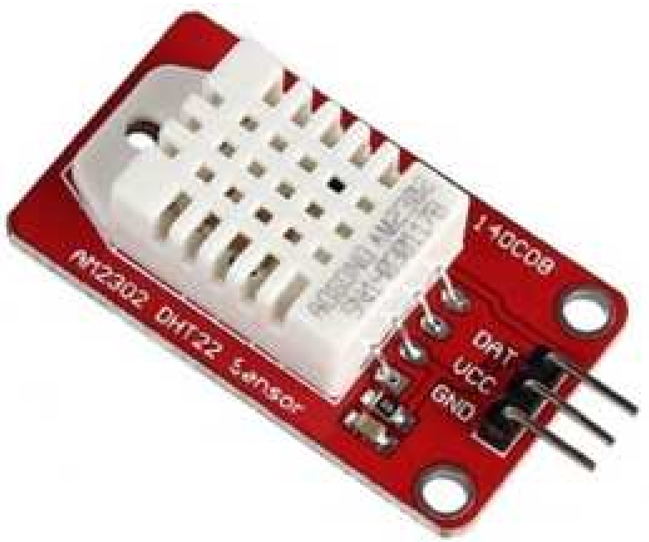

Fig. 10: Temperature and Humidity Sensor (DHT22)

DHT22 depicts in fig. 10 known as the AM2302, is a capacitive humidity sensor and a high precision temperature sensor combined in a single package. It uses specialized digital module acquisition technology as well as temperature and humidity sensor technology. It also has capacitive sensing and high precision temperature measurement elements linked to a high-performance 8-bit microprocessor for overall performance. It has the benefits of outstanding quality, ultra-fast reaction, powerful anti-interference ability, and high-cost performance at a reasonable price [23].

The system design consists of Block Diagrams and Actual Diagram. The sequence diagram and communication diagram both follow the same flow that is depicted in the class diagram to ensure integrity of how the app functions.

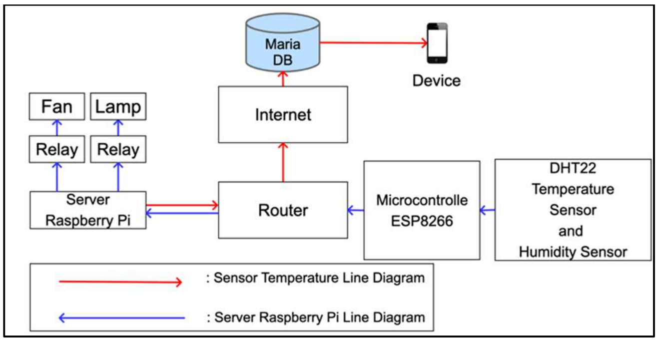

Fig. 11: Block Diagram of Smart poultry farm system

Fig. 11 depicts the design of a temperature and humidity to monitor system for users (the farmer) with a smart poultry farm using the internet of things (IoT). ESP8266 as a microcontroller and Raspberry Pi 4 Model B to receive the data and transfer it to the database. The data output displayed in the smartphone is due to the temperature and humidity result. The system programmed with Python to connect DHT22 as the temperature and humidity sensor into microcontroller ESP8266. Sensor DHT22 will detect temperature and humidity levels, and microcontroller ESP8266 will transfer the result of sensor through the router to Raspberry PI 4 Model B process the data and the output system fan and lamp will detect and automatically turned on if the environment overheated or too cold. The data result will transfer from Raspberry PI 4 Model B into the database through the router and received with Maria DB as the database to collect the result from the sensor, the system in actual diagram using actual devices explained in fig. 12.

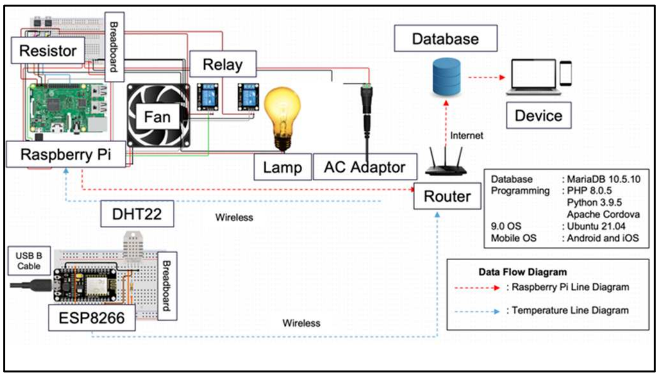

Fig. 12: Actual Diagram for Smart poultry farm system

IV. Finding and Discussion

The transition phase in the Agile Unified Process will elaborate on the sensor testing methods that has taken place during prototyping stages. Types of testing that have been performed is unit testing, integration testing, beta testing and user acceptance testing. The data taken in the test is the result of the DHT22 sensor reading with temperature (°C) and humidity (%). Also, comparing the manual system with a smart poultry farm. The server used in the system to store the data since the system supported 3 Tier of IoT Architecture as a standard IoT system and shown the value of temperature and humidity sensor reading through website with graph.

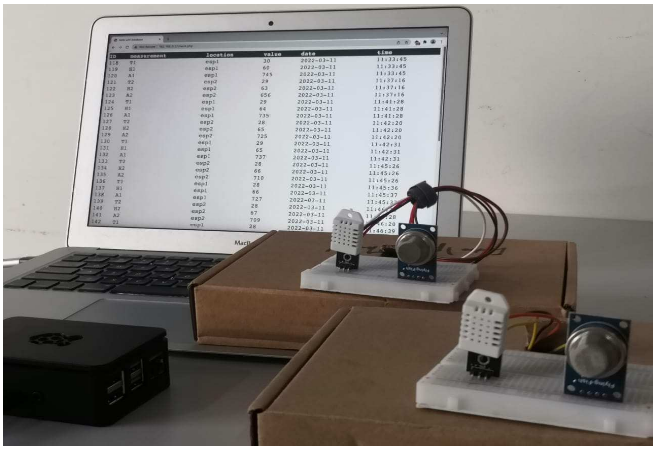

Fig. 13: Smart Poultry Farm system device

Fig. 13 depicts a prototype device for the slave module that has been successfully installed. Slave module consists of a DHT22 temperature and humidity sensor, an ESP8266 microcontroller, and 2.0 USB cable.

Temperature

ID	measurement	location	value	date	time
118	T1	esp1	30C	2022-03-11	11:33:45
121	T2	esp2	29C	2022-03-11	11:37:16
124	T1	esp1	29C	2022-03-11	11:41:28
127	T2	esp2	28C	2022-03-11	11:42:20
130	T1	esp1	29C	2022-03-11	11:42:31
133	T2	esp2	28C	2022-03-11	11:45:26
136	T1	esp1	28C	2022-03-11	11:45:36
139	T2	esp2	28C	2022-03-11	11:46:28
142	T1	esp1	28C	2022-03-11	11:46:39
145	T2	esp2	28C	2022-03-11	11:47:30
148	T1	esp1	28C	2022-03-11	11:47:40
151	T2	esp2	28C	2022-03-11	11:48:32
154	T1	esp1	28C	2022-03-11	11:48:43
157	T2	esp2	28C	2022-03-11	11:49:35
160	T1	esp1	28C	2022-03-11	11:49:46
163	T2	esp2	28C	2022-03-11	11:50:38
166	T1	esp1	28C	2022-03-11	11:50:48

Humidity

ID	measurement	location	value	date	time
119	H1	esp1	60%	2022-03-11	11:33:45
122	H2	esp2	63%	2022-03-11	11:37:16
125	H1	esp1	64%	2022-03-11	11:41:28
128	H2	esp2	65%	2022-03-11	11:42:20
131	H1	esp1	65%	2022-03-11	11:42:31
134	H2	esp2	66%	2022-03-11	11:45:26
137	H1	esp1	66%	2022-03-11	11:45:37
140	H2	esp2	67%	2022-03-11	11:46:28
143	H1	esp1	66%	2022-03-11	11:46:39
146	H2	esp2	67%	2022-03-11	11:47:30
149	H1	esp1	67%	2022-03-11	11:47:41
152	H2	esp2	67%	2022-03-11	11:48:32
155	H1	esp1	67%	2022-03-11	11:48:43
158	H2	esp2	67%	2022-03-11	11:49:35
161	H1	esp1	67%	2022-03-11	11:49:46
164	H2	esp2	67%	2022-03-11	11:50:38
167	H1	esp1	67%	2022-03-11	11:50:48

Fig. 14: Database Result

Database is needed to collect the result from DHT22 sensor readings, with the values of temperature (T1, T2) and humidity (H1, H2). Fig. 14 displays the result analysis of the smart poultry farm system when doing an experiment in the chicken cage with duration in a day.

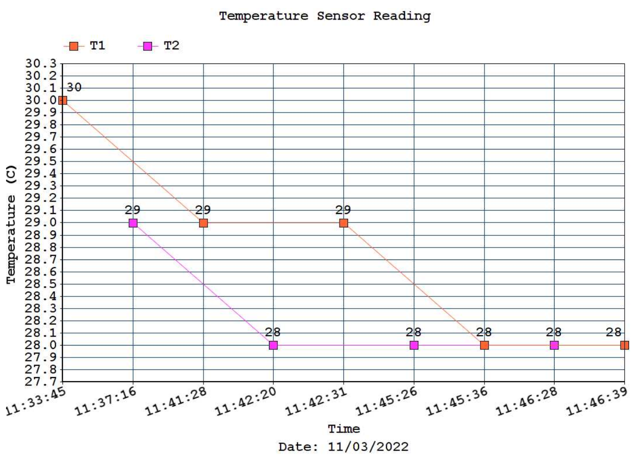

Fig. 15: Temperature parameter result in a day

Fig. 15 depicts the result of temperature level from the experiment that has been run and transferred into the database. The researher has been calculate it using a graph to display result of DHT22 sensor readings every 3 – 4 second for an hour. In the fig. 15 depicts two graphs shown sensor 1 and sensor 2 values which are placed in different rooms but apart from each other, the analysis system shows significant changes in both sensors as seen at 11:42:20 and 11:45:26 have different temperature readings from 28°C - 29°C since the sensors are put in the same place but apart from each other.

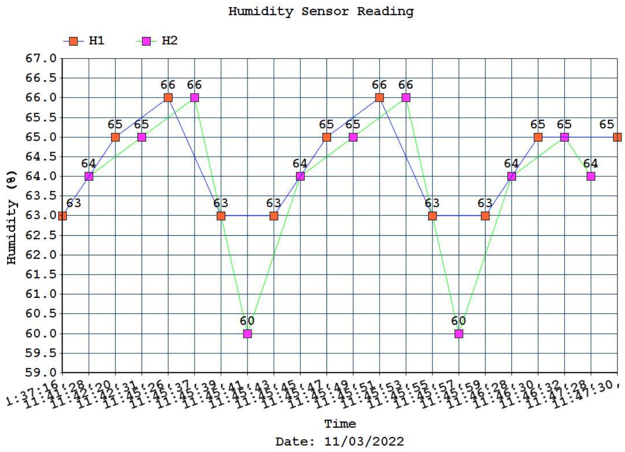

Fig. 16: Humidity parameter result in a day

Also, in fig. 16 depicts two graphs shown sensor 1 and sensor 2 same as temperature values reading which are placed in different rooms but apart from each other, the analysis system shows significant changes in both sensors as seen at 11:45:39 - 11:45:45 have different humidity value readings since the sensors are put in the same place but apart from each other. The result of humidity level from the experiment that has been run and transferred into the database. Similar process with temperature analysis, we calculate it using a graph to display the DHT22 sensor readings every 3 – 4 second for an hour.

V. Conclusion

In this study, an IoT-based design was created that demonstrates how Prototypes and Open-Source Software were utilized to build an ESP8266. The proposed system is for collecting the temperature and humidity in chicken cages in poultry farms. This was accomplished with the assistance of Raspberry Pi and ESP8266 for controlling and monitoring. Real-time monitoring of environmental factors is critical. The frequency of recording is determined by the frequency with which data must be recorded. The system is an integrated Three-Tier IoT architecture, which means that the system provides a server base to store the data as a standard three layers of an IoT architecture. but the smart poultry farm system focuses on information gathering via sensors to analyze sensors accuracy using Raspberry Pi 4 Model B.

As a summary, a traditional poultry farm needs to build a smart poultry farming system to aid small farmers to monitor temperature and humidity levels in chicken cage. An automated control is needed in the poultry farm to adjust the chicken cage temperature automatically and stabilize the temperature and humidity.

In the future, the system will be extended to add more environmental sensors to support the development of smart poultry farming systems such as CO2 values, rainfall values, and smartphones to control the evaporation systems based on threshold values of temperature and humidity. Also, support the basis 3 layer of the IoT architecture to provide the security of storing data via cloud.

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